Jenna M. Kline

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Education

PhD Computer Science & Engineering, **The Ohio State University** 2021-2026

Dissertation: Autonomous Drone Systems for In Situ Animal Ecology: Towards AI-Driven Multi-Modal Ecosystem Monitoring at the Far Edge
Major: Software Systems | ***Minors:*** Artificial Intelligence & Imageomics
Advisors: Dr. Christopher Stewart and Dr. Tanya Berger-Wolf

B.S. Industrial Engineering, *cum laude*, **The Ohio State University** 2014-2018

Honors in Integrated Business and Engineering (IBE)

Research Interests

Cyber-Physical Systems for Adaptive Environmental Sensing at the Far Edge

I build **cyber-physical systems** for autonomous environmental sensing in dynamic, resource-constrained field settings. My research advances **distributed intelligence**, **edge AI**, and **adaptive autonomy** to coordinate heterogeneous sensors and robots, enabling cross-modal systems to adapt what, where, and when they observe. I deploy these systems across ecology, biodiversity monitoring, and coastal and climate resilience, developing the foundations for persistent, adaptive observation of complex environmental systems.

Academic Appointments and Experience

Massachusetts Institute of Technology, Postdoctoral Associate Jun 2026 - present

Department of Civil and Environmental Engineering

Research on coastal resilience, using satellite imagery, drones and computer vision to monitor salt marsh habitats through the Climate Project at MIT, supervised by Dr. Heidi Nepf.

The Ohio State University, Graduate Research Assistant Aug 2021 - May 2026

Research supported by NSF Imageomics (Award 2118240) and ICICLE (Award 2112606).

Visiting PhD Candidate, WildDroneEU 2024-2026

Conducted drone fieldwork in Kenya (2024-2025), research at University of Southern Denmark (2024), University of Bristol (2024), and presented at the WildDrone summer school, hosted by the Max Planck Institute of Animal Behavior (2025).

Los Alamos National Laboratory, Supercomputer Institute Graduate Intern Summer 2023

Completed high performance computing bootcamp. Conducted research on scalable characterization of HPC filesystem trees using GUFI, presented at The International Conference for High Performance Computing, Networking, Storage, and Analysis (SC23) ACM Student Research Competition.

Field Experience

Waquoit Bay National Estuarine Research Reserve, Massachusetts Summer 2026

Team lead and drone pilot. Conducted drone fieldwork monitoring salt-marsh habitats with computer vision, supervising a team of undergraduate researchers. Coastal monitoring project at MIT with Dr. Heidi Nepf.

Ol Pejeta Conservancy, Kenya Winter 2026

Drone pilot. Tested behavior-adaptive drone system for wildlife monitoring [4]. Funded by the Ohio State University Graduate School's Alumni Grants for Graduate Research and Scholarship (AGGRS).

The Wilds Conservation Center, Ohio Summer 2025

Team lead. Collected the SmartWilds multimodal dataset — camera-trap, bioacoustic, and drone data [15].

Ol Pejeta Conservancy, Kenya Winter 2025

Drone pilot. Tested the WildWing system [5] and collected the MMLA dataset [17].

The Wilds Conservation Center, Ohio Summer-Fall 2024

Lead drone pilot. Tested the WildWing system [5] and collected the MMLA dataset [17].

Mpala Research Centre, Kenya Winter 2023

Lead drone pilot. Collected the KABR dataset for in situ animal behavior recognition [20].

Selected Publications

1. **J. Kline**, S. Stevens, D. I. Rubenstein, T. Berger-Wolf, and C. Stewart. "What the Sheep Can Teach the Shepherd: A Vision for Self-Organizing Drone Swarms Informed by Collective Animal Behavior." In: *2026 IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS)*. 2026. To appear. *[Vision paper]*
2. **J. Kline**. "A Maneuver-Indexed Testbed for Context-Aware Adaptive Wildlife Drones." In: *2026 IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C)*. 2026. To appear. *[Peer-reviewed artifact paper]*
3. **J. Kline**, S. Afridi, E. G. A. Rolland, G. Maalouf, L. Laporte-Devylder, et al. "Studying Collective Animal Behaviour with Drones and Computer Vision." *Methods in Ecology and Evolution* 16.10 (2025), pp. 2229–2259. **2025 Robert May Prize**. doi:10.1111/2041-210X.70128.
4. **J. Kline**, R. Katole, T. Berger-Wolf, and C. Stewart. "Edge-Native, Behavior-Adaptive Drone System for Wildlife Monitoring." In: *DroneSys: Workshop on Autonomous Drone Computing Systems and Applications at the ACM/IEEE Symposium on Edge Computing (SEC)*. 2025. Associated work received **First Place in the SEC 2025 Student Research Competition**. doi:10.1145/3769102.3774245.
5. **J. Kline**, A. Zhong, K. Irizarry, C. V. Stewart, C. Stewart, et al. "WildWing: An Open-Source, Autonomous and Affordable UAS for Animal Behaviour Video Monitoring." *Methods in Ecology and Evolution* 17.2 (2025), pp. 291–300. doi:10.1111/2041-210X.70018.
6. **J. Kline**, A. O'Quinn, T. Berger-Wolf, and C. Stewart. "Characterizing and Modeling AI-Driven Animal Ecology Studies at the Edge." In: *2024 IEEE/ACM Symposium on Edge Computing (SEC)*. 2024, pp. 220–233. doi:10.1109/SEC62691.2024.00025.
7. **J. Kline**, C. Stewart, T. Berger-Wolf, M. Ramirez, S. Stevens, et al. "A Framework for Autonomic Computing for In Situ Imageomics." In: *2023 IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS)*. 2023, pp. 11–16. doi:10.1109/ACSOS58161.2023.00018.

Journal Articles

8. G. Maalouf et al., including **J. Kline**. "SORA 2.5-Guided BVLOS UAS for Wildlife Conservation in Kenya: Reducing Friction Between Safety and Field Operations." *Drones* 10.3 (2026). doi:10.3390/drones10030178
9. I. Duporge, M. Kholiavchenko, et al. including **J. Kline**. "BaboonLand Dataset: Tracking Primates in the Wild and Automating Behaviour Recognition from Drone Videos." *International Journal of Computer Vision* 133.9 (2025), pp. 6578–6589. doi:10.1007/s11263-025-02493-5.
10. M. Kholiavchenko, **J. Kline**, M. Kukushkin, O. Brookes, S. Stevens, et al. "Deep Dive into KABR: A Dataset for Understanding Ungulate Behavior from In-Situ Drone Video." *Multimedia Tools and Applications* 84.21 (2025), pp. 24563–24582. doi:10.1007/s11042-024-20512-4.
11. S. Afridi, L. Laporte-Devylder, G. Maalouf, **J. Kline**, S. G. Penny, et al. "Impact of Drone Disturbances on Wildlife: A Review." *Drones* 9.4 (2025), article 311. doi:10.3390/drones9040311.

Peer-Reviewed Conference and Workshop Publications

12. B. Pillai, V. Viswapriyan, C. Stewart, T. Berger-Wolf, and **J. Kline**. "Cross-Modal Corroboration for Annotation-Free Wildlife Monitoring." In: *CV4Animals Workshop at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026. arXiv:2606.21613
13. C. Sun, T. Berger-Wolf, and **J. Kline**. "Autonomous UAV Navigation for Individual Wildlife Re-Identification." In: *CV4Animals Workshop at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2026. arXiv:2606.31772
14. A. O'Quinn, C. Snedeker, S. Zhang, and **J. Kline**. "Environment-Aware Dynamic Pruning for Pipelined Edge Inference." In: *2025 IEEE International Conference on Edge Computing and Communications (EDGE)*. 2025, pp. 83–89. doi:10.1109/EDGE67623.2025.00018
15. **J. Kline**, A. Potlapally, B. Pillai, T. Wani, R. Katole, et al. "SmartWilds: Multimodal Wildlife Monitoring Dataset." In: *Third Workshop on Imageomics at NeurIPS*. 2025. arXiv:2509.18894.
16. S. Stevens, S. M. Rayeed, and **J. Kline**. "Mind the (Data) Gap: Evaluating Vision Systems in Small Data Applications." In: *Third Workshop on Imageomics at NeurIPS*. 2025. arXiv:2504.06486.
17. **J. Kline**, S. Stevens, G. Maalouf, C. R. Saint-Jean, D. N. Ngoc, et al. "MMLA: Multi-Environment, Multi-Species, Low-Altitude Aerial Footage Dataset." In: *CV4Animals Workshop at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025. arXiv:2504.07744.
18. N. N. Dat, T. Richardson, M. Watson, K. Meier, **J. Kline**, et al. "WildLive: Near Real-Time Visual Wildlife Tracking Onboard UAVs." In: *CV4Animals Workshop at the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. 2025. arXiv:2504.10165.

19. **J. Kline**. "An Adaptive Autonomous Aerial System for Dynamic Field Animal Ecology Studies." In: *2024 IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C)*. 2024, pp. 164–166. doi:10.1109/ACSOS-C63493.2024.00051 *[PhD Forum extended abstract]*
20. M. Kholiavchenko, **J. Kline**, M. Ramirez, S. Stevens, A. Sheets, et al. "KABR: In-Situ Dataset for Kenyan Animal Behavior Recognition from Drone Videos." In: *IEEE/CVF Winter Conference on Applications of Computer Vision Workshops (WACVW)*. 2024, pp. 31–40. doi:10.1109/WACVW60836.2024.00011.
21. **J. Kline**, M. Kholiavchenko, O. Brookes, T. Berger-Wolf, C. V. Stewart, et al. "Integrating Biological Data into Autonomous Remote Sensing Systems for In Situ Imageomics: A Case Study for Kenyan Animal Behavior Sensing with Unmanned Aerial Vehicles." In: *First Workshop on Imageomics at AAAI*. 2024. arXiv:2407.16864.
22. D. Grushchak, **J. Kline**, D. Pianini, N. Farabegoli, G. Aguzzi, et al. "Decentralized Multi-Drone Coordination for Wildlife Video Acquisition." In: *2024 IEEE International Conference on Autonomic Computing and Self-Organizing Systems (ACSOS)*. 2024, pp. 31–40. **ACSOS 2024 Best Poster Award**. doi:10.1109/ACSOS61780.2024.00021.
23. D. Grushchak, **J. Kline**, D. Pianini, and N. Farabegoli. "An Agent-Based Model of Directional Multi-Herds." In: *2024 IEEE International Conference on Autonomic Computing and Self-Organizing Systems Companion (ACSOS-C)*. 2024, pp. 15–20. doi:10.1109/ACSOS-C63493.2024.00023.
24. E. Rolland, K. A. R. Grøntved, L. Laporte-Devylder, **J. Kline**, U. P. S. Lundquist, et al. "Drone Swarms for Animal Monitoring: A Method for Collecting High-Quality Multi-Perspective Data." In: *15th International Micro Air Vehicle Conference (IMAV)*. 2024, pp. 316–323.
25. **J. Kline**, J. Lee, and R. Davis. "Seeing the Trees for the Forest: Describing HPC Filesystem Trees with the Grand Unified File-Index (GUFI)." In: *ACM Student Research Competition at the International Conference for High Performance Computing, Networking, Storage, and Analysis (SC23)*. Denver, CO, 2023. *[Extended abstract; presented as poster]*

Preprints and Articles Under Review

26. **J. Kline** et al. "Where to Trust Automation in Remote-Sensing-Based Animal Behavior Monitoring: An Empirical Evaluation Using Drone and Camera-Trap Video." 2026.
27. **J. Kline**, A. Potlapally, H. Subramoni, T. Berger-Wolf, and C. Stewart. "A Blueprint for Cross-Modal Coordination at the Far Edge." 2026.
28. **J. Kline**, K. Meier, V. Shukla, E. G. A. Rolland, E. Iannino, et al. "FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets." 2026. arXiv:2606.00355.
29. L. J. Pollock, P. H. P. Braga, C. R. Florian, K. Hébert, **J. Kline**, et al.. "Putting the 'Adaptive' in Adaptive Monitoring: From Fast Data to Meaningful Ecological Change.", 2026. EcoEvoRxiv preprint. doi:10.32942/X2S94C.
30. K. Meier, E. G. A. Rolland, E. Iannino, M. Simpson, **J. Kline**, et al. "Simulated and Field-Based Error Characterization of Animal Geolocalisation and Relative Positioning via Commercial Drones.", 2026.
31. G. Aguzzi, M. Baiardi, N. Farabegoli, D. Grushchak, **J. Kline**, D. Pianini, C. Stewart, and M. Viroli. "Simulation-Grounded Decentralized Multi-UAV Coordination for Scalable In-Situ Imageomics of Wildlife Herds.", 2026.
32. V. Shukla, K. Meier, L. Laporte-Devylder, C. R. Saint-Jean, **J. Kline**, et al. "WildBox: A Dataset and Benchmark for Aerial Monocular 3D Detection of African Savanna Wildlife." Submitted, 2026. arXiv:2606.21309.
33. **J. Kline**, M. Kholiavchenko, S. Stevens, N. van Tiel, A. Zhong, et al. "kabr-tools: Automated Framework for Multi-Species Behavioral Monitoring." arXiv preprint, 2025. arXiv:2510.02030.

Invited Lectures and Presentations

34. **J. Kline** et al. "FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets." Accepted abstract, Biodiversity Information Standards Annual Conference (TDWG), Oslo, Norway, September 2026.
35. E. Sokol et al., including **J. Kline**[*]. "A Metadata-First FAIR4AI Checklist and Automated Evaluation Agent for Biodiversity Datasets." Accepted abstract, TDWG, Oslo, Norway, September 2026. [*]Presenting author.
36. **J. Kline** et al. "Cross-Modal Coordination for Biodiversity Monitoring at Scale: Lessons from SmartWilds." Accepted abstract, TDWG, Oslo, Norway, September 2026.
37. **J. Kline** et al. "Digital Twins as System-Optimization Tools: Provisioning Edge and Cloud Infrastructure for Biodiversity Monitoring at Scale." Accepted abstract, TDWG, Oslo, Norway, September 2026.
38. **J. Kline** et al. "kabr-tools: A Modular Workflow for Automated Video-Based Behavioral Monitoring Across the Autonomy Spectrum." Accepted abstract, TDWG, Oslo, Norway, September 2026.
39. **J. Kline**. "Lessons from FAIR^2 Drones: Toward AI-Ready Multimodal, Multispatial Sensor Data for Ecology." Invited symposium talk, "Building an AI-Ready Ecology and Biodiversity Data Infrastructure for Science and Action," Ecological Society of America (ESA) Annual Meeting, Salt Lake City, UT, July 2026. Scheduled.
40. **J. Kline**. Panelist, "AI for Ecology: Accelerating Discoveries, Reducing Uncertainties, and Scaling Solutions." Special Session SS23, Ecological Society of America (ESA) Annual Meeting, Salt Lake City, UT, July 2026. Scheduled.

41. **J. Kline**. "Autonomous Drones for Animal Studies with Edge AI." Invited talk, Global Conservation Tech & Drone Forum, Nairobi, Kenya, 2026.
42. **J. Kline** et al. "FAIR^2 Drones: An AI-Ready Standard for Cross-Domain Wildlife Drone Datasets." Poster presentation, FARR Workshop: FAIR in ML, AI Readiness and Reproducibility, Washington, DC, 2026.
43. **J. Kline**, R. Katole, and C. Stewart. "An Edge-Native Approach to Behavior-Adaptive Navigation in Drone Systems." In: *Proceedings of the Tenth ACM/IEEE Symposium on Edge Computing (SEC)*. 2025, pp. 1–3. **SEC 2025 Student Research Competition — First Place**. doi:10.1145/3769102.377437. *[Extended abstract; presented as poster]*
44. P. Covey, **J. Kline**, and C. Stewart. "An Edge-to-Cloud Framework for Vigilance-Adaptive Drones." In: *Proceedings of the Tenth ACM/IEEE Symposium on Edge Computing (SEC)*. 2025, pp. 1–3. doi:10.1145/3769102.3774384. *[Extended abstract; presented as poster]*
45. **J. Kline**. "Autonomous AI-Driven Environmental Sensing Systems." Poster presentation, MIT and Boston University Rising Stars in EECS Workshop, 2025.
46. **J. Kline**. "How Do Drones Fit into Multimodal Sensing Networks?" Poster presentation, WildDrone Summer School, Max Planck Institute of Animal Behavior, Konstanz, Germany, 2025.
47. **J. Kline**. "Autonomous, Adaptive Vision-Based Remote Sensing System for Dynamic Field Animal Ecology Studies." Oral presentation, Edward F. Hayes Advanced Research Forum, The Ohio State University, 2025. **First Place, Engineering Oral Presentation**.
48. **J. Kline**. "Drones and AI in Field Animal Ecology: Integrating Biology, Animal Science, and Technology." Invited lecture, Fridays at Findlay, University of Findlay, 2025.
49. **J. Kline**. "From Ohio to Kenya: Autonomous Drones in Field Ecology and Conservation." Invited lecture, Conservation Science Symposium, Muskingum University and The Wilds Conservation Center, 2025.
50. **J. Kline**. "Autonomous Drones for Collective Wildlife Behavior." Invited lecture, WildDrone Seminar Series, 2025.
51. **J. Kline**. "Adaptive, Autonomous Aerial Systems for Dynamic Field Animal Ecology Studies." Invited lecture, University of Bologna, Cesena, Italy, 2024.
52. **J. Kline**. "Individual Identification of Zebras with Autonomous UAV Swarms." Poster presentation, TDAI Interdisciplinary Research Fall Forum, The Ohio State University, 2024. **Best Poster Award**.
53. **J. Kline**. "Intelligent Cyberinfrastructure for Remote Sensing with Autonomous Aerial Robotics." Poster presentation, TDAI Interdisciplinary Research Fall Forum, The Ohio State University, 2023.
54. **J. Kline**. "Interdisciplinary Applications of Autonomous Unmanned Aerial Vehicles." Poster presentation, STARS Celebration at Tapia and Ohio Celebration of Women in Computing, 2023.
55. **J. Kline**. "Autonomous UAV Missions for Studying Wildlife Behavior: A Case Study for the Individual Identification of Zebras." Poster presentation, Midwest Machine Learning Symposium (MMLS), Chicago, IL, 2023.
56. **J. Kline**. "Individual Identification of Zebras with Autonomous UAV Swarms." Poster presentation, CRA-WP Grad Cohort for Women, San Francisco, CA, 2023.
57. **J. Kline**. "PhD Forum: Edge Computing for Software-Defined Cartography." PhD Forum and extended abstract, ACM/IEEE Symposium on Edge Computing (SEC), Seattle, WA, 2022.

Artifacts: Models, Datasets, and Software

58. **J. Kline** et al. *kabr-tools: Automated Framework for Multi-Species Behavioral Monitoring.* Open-source software for analyzing animal behavior from in situ video, 2025–present. [Code].
59. **J. Kline**. *A Maneuver-Indexed Evaluation Framework for Context-Aware Adaptive Wildlife Drones.* Interactive research demo, Hugging Face Spaces, 2026. Companion artifact for the ACSOS-C 2026 paper. [Demo].
60. **J. Kline** et al. *WildWing: Open-Source Autonomous UAS for Animal Behaviour Monitoring.* Open-source hardware and software system. [Code].
61. **KABR: In Situ Kenyan Animal Behavior Recognition Data and Model Collection.** Imageomics, Hugging Face, 2024–present. Drone videos, behavior annotations, synchronized telemetry, worked examples, and models. *Role:* data collection, annotation, analysis, telemetry integration, and public release. [Collection].
62. **MMLA: Multi-Environment, Multi-Species, Low-Altitude Aerial Data and Model Collection.** Imageomics, Hugging Face, 2025–present. Multi-site detection datasets, pose annotations, deployment data, and trained detection and pose models. *Role:* data collection, annotation, analysis, curation, and public release. [Collection].
63. **SmartWilds: Multimodal Wildlife Monitoring Data Collection.** Imageomics, Hugging Face, 2025–present. Camera-trap, bioacoustic, and drone monitoring resources collected at The Wilds Conservation Center. *Role:* project lead; led study design, field data collection, annotation, analysis, multimodal coordination, and public release. [Collection].

Grants and Awards

Presidential Fellow: The Ohio State University Presidential Fellowship 2025–2026
Robert May Prize: Best early-career-author paper, British Ecological Society [3] 2025
Student Research Competition - First Place: IEEE/ACM Symposium on Edge Computing [43] 2025
Departmental Research Award: Outstanding Graduate Research in CSE, OSU 2025
Hayes Forum - First Place: Hayes Advanced Research Forum, OSU [47] 2025
Rising Stars in EECS: MIT & Boston University Rising Stars Workshop 2025
WiscProf Workshop: Future Faculty in Engineering, Univ. of Wisconsin–Madison 2025
Research Grant: OSU Alumni Grant for Graduate Research and Scholarship 2025
NSF NAIRR Pilot Allocation: Digital Twins for Ecological Research (NAIRR250091) 2025
ACSOS Best Poster Award: Autonomic Computing & Self-Organizing Systems (ACSOS) [22] 2024
TDAI Best Poster Award: OSU Translational Data Analytics Institute Fall Forum [52] 2024
Travel Grant: ACM/IEEE Symposium on Edge Computing 2022 & 2024
CRA-WP Grad Cohort: Computing Research Association Grad Cohort for Women [56] 2023
Grace Hopper Scholar: The Ohio State University ACM-W 2022

Student Supervision and Teaching Experience

Student Supervision

Massachusetts Institute of Technology (2026–present). Supervised two summer undergraduate researchers (UROP) on a salt-marsh monitoring study using drones and computer vision.

The Ohio State University (2022–2026). Supervised five undergraduate researchers, including a team conducting fieldwork at The Wilds Conservation Center, and led a field-campaign team of undergraduate, master's, and PhD students. Student-led projects were published at CV4Animals at CVPR 2026 [12, 13] and SEC 2025 [44].

University of Bologna (2024). Co-supervised Denys Grushchak's master's thesis, *Herd Monitoring with Autonomous Drones: A Decentralized k-Coverage-Inspired Approach*, with Dr. Danilo Pianini; the thesis resulted in two ACSOS publications [22, 23].

Teaching Experience

All courses taught at The Ohio State University, Department of Computer Science and Engineering.

Lab Instructor — *CSE 2111: Modeling and Problem Solving with Spreadsheets and Databases.* Undergraduate required course. Led two 30-student lab sections twice weekly and supervised a team of undergraduate graders. Topics: spreadsheet and database modeling, programming concepts, business applications.

Graduate Teaching Assistant — *CSE 3244: Data Management in the Cloud.* Senior-level elective. Sole graduate teaching assistant responsible for grading and supervising lab assignments and the final project. Topics: data organization on cloud computing architectures, B-tree and hash-based indexing, query optimization fundamentals, data partitioning, and distributed task scheduling.

Service

Community Building Co-chair: ACSOS 2026; organized community-building seminar series 2026-present
Workshop Organizer & Lead Facilitator: Edge AI for Conservation workshop, ICTC, Lima, Peru 2026
Group Co-lead: Edge Computing for Conservation Group, WILDLABS Conservation Technology Network 2025–present
Workshop Organizer: Imageomics Image Datapalooza Hackathon 2023
Volunteer: Code I/O coding camp, The Ohio State University ACM-W Chapter 2023
Peer Mentor: Graduate Student Alumni and Peer Mentoring Program, The Ohio State University 2022–2024

Journal Reviewer: Ecological Informatics, Scientific Reports, International Journal of Computer Vision (IJCV), Ecological Modelling, Methods in Ecology and Evolution, Ecology and Evolution, IET Computer Vision, PLOS One, Marine Mammal Science, Global Ecology and Conservation, MethodsX

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